

1. Administrative & Study Identification

Full Study Title: A placebo-controlled, patient and investigator blinded, randomized parallel cohort study to assess pharmacodynamics, pharmacokinetics, safety, tolerability and preliminary clinical efficacy of VAY736 and CFZ533 in patients with systemic lupus erythematosus (SLE) **Short Title/Acronym:** Study the Efficacy and Safety of VAY736 and CFZ533 in SLE Patients **Protocol Number:** CVAY736X2208 **NCT Number:** NCT03656562 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** VAY736 (ianalumab) and CFZ533 (iscalimab) **Phase of Development:** Phase 2 **Medical Monitor:** Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety, tolerability, pharmacokinetics, and therapeutic efficacy of treatment with either VAY736 (ianalumab) or CFZ533 (iscalimab) in patients with systemic lupus erythematosus (SLE), measured primarily by the proportion of patients achieving an SLE Responder Index-4 (SRI-4) response with sustained steroid reduction. **Secondary Objectives:** To assess pharmacodynamics (reduction in B-cell subsets), achievement of Lupus Low Disease Activity State (LLDAS), reduction in SLEDAI-2K scores, and incidence of moderate or severe SLE flares.

2.2 Background & Rationale To address the need for targeted, steroid-sparing therapies in moderate-to-severe SLE patients who continue to experience significant symptoms despite standard treatments. Ianalumab (VAY736) is an afucosylated monoclonal antibody targeting the B-cell activating factor receptor (BAFFR), which works to deplete B cells via enhanced antibody-dependent cellular cytotoxicity and concurrent BAFF:BAFFR blockade. This offers a novel mechanism to potentially reduce disease activity and the long-term corticosteroid burden. CFZ533 (iscalimab) provides an alternative pathway investigation.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind (Patient and Investigator blinded) **Randomization:** Randomized, parallel cohort (1:1)

3.2 Planned Enrollment & Duration Target N: 107 participants
Number of Sites: 31 global locations **Estimated Study Start:** 19-Dec-2018 **Estimated Primary Completion:** 27-Jul-2022

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adult men or women, 18 to 75 years of age. 2. Diagnosed with SLE for at least 6 months prior to screening, fulfilling ≥ 4 of the 11 ACR 1997 classification criteria. 3. Elevated serum titers at screening of ANA ($\geq 1:80$) and specific autoantibodies (e.g., anti-dsDNA, anti-Ro, anti-La, anti-Sm). 4. SLEDAI-2K score of ≥ 6 at screening. 5. Currently receiving corticosteroids, anti-malarial, thalidomide, and/or another DMARD on a stable dose.

4.2 Exclusion Criteria 1. Active viral, bacterial, or other infections at the time of screening or enrollment, including HIV, or chronic HBV/HCV. 2. Presence of WHO Class III-IV renal involvement or severe lupus kidney disease (defined by proteinuria > 6 g/day or equivalent). 3. Severe organ dysfunction or life-threatening disease; ECOG performance status > 1 . 4. History of receiving B-cell depleting therapies within 52 weeks prior to screening, or recent use of biologics like belimumab or abatacept within 24 weeks.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **VAY736 Cohort:** 300 mg (administered as two 150 mg doses) every 4 weeks. * **CFZ533 Cohort:** Administered every 4 weeks based on body weight (10 mg/kg for ≥ 50 kg; 13 mg/kg for < 50 kg). * **Placebo:** Matching placebo every 4 weeks. **Route of Administration:** Subcutaneous (s.c.) for VAY736; Intravenous (i.v.) for CFZ533. **Duration of Treatment:** 28-week blinded treatment phase followed by an open-label active treatment phase up to Week 49, with safety follow-up extending to Week 69.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care (SoC) for SLE, including stable doses of corticosteroids, anti-malarials, and specific oral DMARDs. Corticosteroid tapering was required per protocol responder criteria. **Prohibited:** Live/attenuated vaccines within 2 months before first dosing, concurrent biologic therapies, and non-protocol B-cell depleting agents.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to 28 Days Informed Consent, Medical History, SLEDAI-2K scoring, Autoantibody & Viral Labs

Baseline | Day 0 | Randomization, First Dose (Blinded Phase), Efficacy/Safety baseline | | **Interim** | Weeks 4-28 | Q4W Dosing (in-person), Clinical Assessments, Corticosteroid Tapering Tracking | | **Open-Label** | Weeks 29-49 | Switch to Open-Label VAY736/CFZ533, continued Q4W Dosing | | **End of Study** | Week 69 | Safety Follow-Up, Efficacy Evaluation, B-cell recovery tracking |

(Note: Participants failing to achieve B-cell recovery by Week 69 enter a secondary follow-up period until B-cell count reaches ≥ 50 cells/ μ L or at least 80% of baseline)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Proportion of patients achieving an SLE Responder Index-4 (SRI-4) response combined with sustained corticosteroid reduction (tapering prednisone ≤ 5 mg/d or $\leq$ baseline dose, whichever is lower, by Week 16 and sustained to Week 28). **Measurement Method:** Clinical scoring assessments (SLEDAI-2K, BILAG, PGA) and medication log verification.

7.2 Secondary & Exploratory Endpoints * Reduction in B-cell subsets (assessed by flow cytometry) and B-cell-related gene signatures. * Achievement of Lupus Low Disease Activity State (LLDAS). * Incidence of moderate or severe SLE flares. * Pharmacokinetics (PK) and Pharmacodynamics (PD) of VAY736 and CFZ533.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring through physical exams, lab tests, and AE reporting at every Q4W visit.

Serious Adverse Events (SAEs): Standard reporting timelines applied. Interim analyses monitored for thromboembolic risks, severe infections, and prolonged B-cell depletion. **Data**

Monitoring Committee (DMC): Utilized to review interim safety data and progression between the blinded and open-label phases.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy outcomes; Safety Set for all participants receiving at least one dose of the investigational product.

Sample Size Justification: $N=107$ designed to provide sufficient exploratory power to detect clinical responses (SRI-4) and biomarker modulation between active treatment cohorts and placebo.

Handling of Missing Data: Handled per protocol; standard imputation rules (e.g., Non-Responder Imputation) typically applied for binary composite endpoints like SRI-4 if patients withdrew or required rescue medication.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and centralized remote monitoring to ensure protocol adherence, accurate administration of weight-based i.v./fixed s.c. treatments, and valid clinical scoring.

Audit Trail: Validated Electronic Data Capture (EDC) system with full audit trail capabilities for DCF processes.

11. Study Identifiers & Governance

Primary ID: CVAY736X2208 **Registry Number:** NCT03656562 **Sponsor/Lead Organization:** Novartis Pharmaceuticals

Study Title: Study the Efficacy and Safety of VAY736 and CFZ533 in SLE Patients

Official Regulatory Title: A placebo-controlled, patient and investigator blinded, randomized parallel cohort study to assess pharmacodynamics, pharmacokinetics, safety, tolerability and preliminary clinical efficacy of VAY736 and CFZ533 in patients with systemic lupus erythematosus (SLE)

12. Clinical Framework (SLE Specific)

Primary Condition: Systemic Lupus Erythematosus (SLE)

Diagnosis Threshold: ≥ 4 of 11 ACR 1997 classification criteria,

SLEDAI-2K ≥ 6 , and ANA $\geq 1:80$

Medical Methodology: Targeted biologic therapy (BAFF-R targeting or CD40 pathway) alongside Standard of Care (SoC) **Optimization Target:** SRI-4 response with sustained corticosteroid reduction

13. Study Procedures & Methodology

Management Pathway: Double-blind biological intervention with protocol-driven corticosteroid tapering

Education Protocol (HCP): Site initiation visits and standardized training on complex SLE

composite scoring (SLEDAI-2K, BILAG)

Education Protocol (Patient): Informed consent detailing the risks of

immunosuppression and B-cell depletion management

Quality Audit Mechanism: Centralized review of clinical scores and centralized laboratory-based flow cytometry for B-cell depletion confirmation

14. Observation & Follow-Up Schedule

Total Duration: Up to 69 weeks (plus secondary follow-up for B-cell recovery if needed)

Onsite Visit Cadence: Every 4 weeks (Q4W) during active treatment up to Week 49

Remote Monitoring Frequency: Continuous safety monitoring

Checklist Review Interval: Regular intervals aligned with Q4W dosing visits

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					